

Student Productivity and Academic Performance **Analytics**

A Data-Driven Analysis of Factors Affecting Student Productivity
and Academic Performance

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Project Overview

Objective:

The objective of this project is to analyze the relationship between study habits, sleep patterns, screen time, stress levels, attendance, productivity, and academic performance among engineering students.

Tools Used:

- Google Forms
- Microsoft Excel
- Power BI
- Microsoft PowerPoint

Data Collection:

Data was collected using a Google Form survey distributed among engineering students from different branches and academic years in IITs and NITs.

Total Responses: 66

Methodology

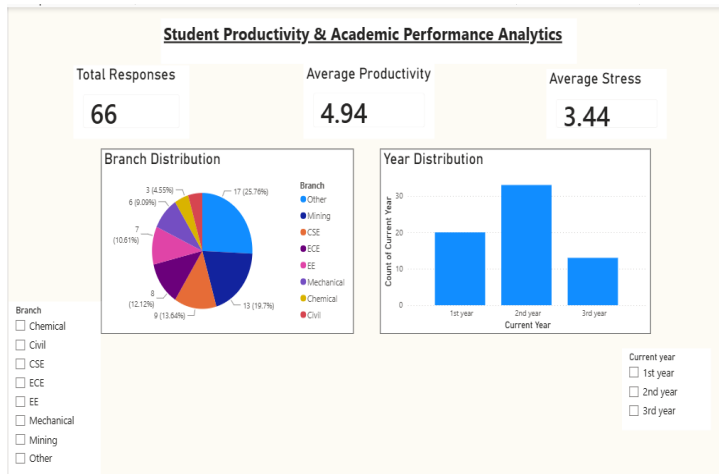
Data Collection Method:

A structured survey was conducted using Google Forms to collect information regarding students' study habits, screen time, sleep duration, stress levels, attendance, productivity and academic performance.

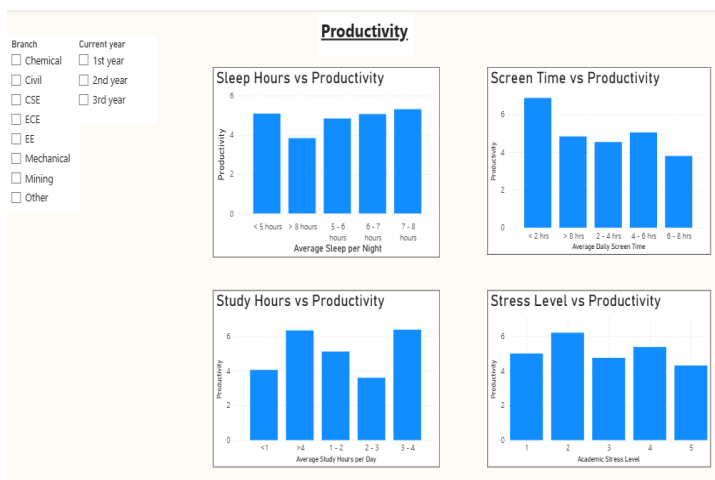
Survey Categories:

- Basic Information
- CGPA
- Study Hours
- Attendance
- Sleep Duration
- Screen Time
- Stress Levels
- Extracurriculars

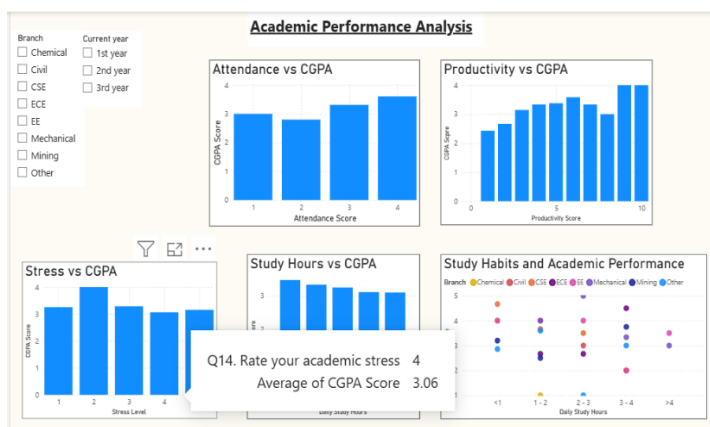
Power BI Dashboard



The Overview Dashboard provides a summary of respondent demographics and average levels of productivity and stress.



The Productivity Analysis page studies the relationship between sleep, screen time, study hours, stress levels and productivity.



This page explores the factors affecting performance using attendance, productivity, stress levels and study hours.

Key Insights and Recommendations

Key Findings:

1. Students with higher sleep scores demonstrated higher productivity.
2. Increased screen time was associated with lower productivity.
3. Attendance showed a positive relationship with CGPA.
4. Higher productivity scores correlated with better academic performance.
5. High stress levels negatively affected productivity and CGPA.

Key Recommendations:

1. Encourage healthy sleep habits.
2. Promote digital wellness and responsible screen usage.
3. Conduct workshops on time management and study planning.
4. Provide stress management support.
5. Improve student engagement and attendance.

Conclusion

This study demonstrates that sleep habits, screen time, stress levels, attendance and study patterns significantly influence student productivity and academic performance. Data-driven interventions can help students improve both academic and personal outcomes.